

DM Module Lab Supplement

STATION: PERITONEUM 1

The peritoneal cavity is the largest serous cavity in the body and is also the most complex. However, it retains the basic features of these cavities with some additions and modifications.

1. The greater sac (): This term refers to the general peritoneal cavity which is automatically opened once the abdominal cavity is entered. It is a potential space between the parietal peritoneum lining the abdominal walls and the visceral layer covering the various viscera.
2. The general peritoneal cavity is subdivided into two main compartments for descriptive purposes: one is above the transverse colon and is known as the supracolic compartment (), containing structures such as the liver and stomach.
3. The other compartment lies below the transverse colon and is termed the infracolic compartment (). It contains among other structures the small intestine.
4. Any two folds of the peritoneum together are called a ligament. For example, check the falciform ligament () that connects the liver to the anterior abdominal wall.
5. There are other ligaments in the peritoneal cavity that do not resemble the above example. These include the median, medial, and lateral umbilical ligaments () found on the posterior aspect of the anterior abdominal wall. They represent peritoneal folds over certain structures.
6. A mesentery is also a double fold of peritoneum that connects an organ to the posterior abdominal wall (usually). An example is the mesentery of the small intestine ().
7. When a double fold of peritoneum is connected to the stomach it is usually called omentum. The greater omentum () however is a four-layer peritoneal fold (two folds of two layers together) hanging from the stomach.
8. The lesser omentum () is a two-layer peritoneal fold attaching the stomach to the liver.

STATION: PERITONEUM 2

Some organs have a complete peritoneal covering (technically no organ is completely covered by peritoneum) and are called intraperitoneal. Some have a partial peritoneal covering on one surface only and are called retroperitoneal or extraperitoneal.

1. The small intestine is an example of an intraperitoneal organ (). Note that it is mobile on its mesentery (). Name one other intraperitoneal organ.
2. The descending colon is a retroperitoneal organ (). Note that it is fixed in its place. Name one other extraperitoneal organ.

The way the organs are arranged creates recesses, pouches, and gutters. These are of great clinical importance. These include:

1. The lesser sac (). This is the largest peritoneal recess, and it is situated behind the stomach. Examine its opening known as the epiploic foramen (of Winslow). Examine its cavity from the opening made into the mesentery of the transverse colon. Name and identify the borders of the foramen of Winslow. Identify the lesser sac and name the organs lying beneath its peritoneum, also called the stomach bed.
2. Place your left index finger into the foramen of Winslow, from greater into lesser sac, and pinch your thumb to your index finger. You have now performed the 'Pringle maneuver'. What ligament is between your fingers and what structures are compressed? What is the significance of performing this maneuver?
3. Around the liver are several recesses. The right and left subphrenic spaces () are between the liver and diaphragm on either side of the falciform ligament. Below the right lobe of the liver is the right subhepatic space (), also known as the hepatorenal pouch of Morrison. The left subhepatic space is the lesser sac.
4. Around the duodenojejunal (DJ) junction can be found sometimes up to three recesses which are named according to their position as superior, inferior, and left paraduodenal recesses ().
5. Behind the caecum is a retrocecal recess ().
6. Occasionally behind the mesentery of the sigmoid can be found the recess of the sigmoid mesocolon ().
7. A modified form of a recess is a gutter. There are four gutters, named right and left paracolic gutters, found on either side of the ascending and descending colon (). Note that the right paracolic gutter of the ascending colon connects the right subhepatic and right subphrenic spaces to the pelvic cavity below (). Clinically it is the most important of the gutters.

STATION: STOMACH

The stomach is the most dilated part of the foregut. Study the provided specimen and identify the following:

1. The cardiac end () to which the esophagus is attached
2. The pyloric end () to which is attached the duodenum. Identify the gastroduodenal artery ()
3. The dome-shaped fundus () supplied by the short gastric arteries ()
4. The region of the stomach body ()
5. The region of the pyloric antrum () together with the thick pylorus ().
6. The lesser curvature () along which traveling the right and left gastric vessels () and branches of the vagus nerves (). The left gastric artery comes from the direction of the esophagus, while the right gastric artery comes from the direction of the pylorus.
7. The greater curvature (), which has the greater omentum attached to it as well as the right and left gastroepiploic vessels running along it ().

8. The interior of the stomach is thrown into folds called rugae (). These folds disappear when the organ is distended.

STATION: SMALL INTESTINES

The word intestine is used to describe the parts of the gastrointestinal tract (GIT) that are distal to the stomach up to the anal canal. They have in common their hollow tubular structure. The small intestine is longer but smaller in caliber than the large intestine. The small intestine starts at the gastroduodenal junction and continues to the ileocecal junction. The first part of the small intestine is the duodenum. This is followed by the jejunum and then the ileum. In the body specimen verify the following structures in the stations: duodenum, jejunum, and ileum.

STATION: SMALL INTESTINE - DUODENUM

The duodenum is the first part of the small intestine, and it is partly in the foregut and partly in the midgut. Identify the following:

1. The first part is intraperitoneal () and is sometimes called the duodenal cap. Note that internally it is smooth, unlike the remainder of the organ ().
2. The second part () is the longest and is also called the descending part. Inside are a number of permanent mucosal folds () and two prominent papillae, the major and minor duodenal papillae (). The duodenal papillae are found on the posteromedial wall of the duodenum with the major being larger and situated about 2 cm below the smaller minor papilla. The second, third, and fourth parts of the small intestine are retroperitoneal.
3. The third part is the horizontal part (), which has no specific features.
4. The fourth part, or the ascending part (), is the last portion of the duodenum. Its junction with the jejunum is marked by a ligament called the ligament of Treitz.

Note that the duodenum forms a C shape that fits the head of the pancreas and is thus in close contact with all the parts of the duodenum (). The mucosal folds that appear in the duodenum are known as plicae circularis. These are permanent folds of the mucous membrane of the small intestine and are present throughout its length but with differing densities by location. They increase the surface area available for absorption. Name and identify the arteries supplying the four parts of the duodenum ().

STATION: SMALL INTESTINE - JEJUNUM

In the body specimen in front of you try to verify the following facts:

1. The small intestine from the (DJ) flexure to the ileocecal junction is suspended from a mesentery that is attached obliquely to the posterior abdominal wall (). It starts just below the transverse mesocolon to the left side of the midline and continues straight down in an oblique direction towards the right iliac fossa. The mesentery is much shorter than the total length of the small intestine, which is

why you will find the small intestine coiled up in the middle of the abdomen. However, the exact location of any of its parts is highly variable due to the mobility conferred upon it by its mesentery. The attachment of the mesentery to the intestine is the mesenteric border of the intestine. The antimesenteric border lies opposite the mesenteric attachment ().

2. Between the right side of the small intestinal mesentery and the left side of the descending colon is the left paracolic gutter (). If you follow it with your hand you will find that it is closed proximally by the attachment of the transverse mesocolon and distally by the small intestinal mesentery ().
3. The right paracolic gutter of the descending colon lies between the right side of the ascending colon and the left side of the small intestine (). It is closed proximally by the transverse mesocolon and leads distally to the pelvic cavity ().
4. The jejunum usually lies upwards and towards the right side of the abdominal cavity due to the way in which the mesentery is attached ().
5. Feel the wall between your index and thumb. You will find that it is thick and gives a feeling that it is made up of two coats ().
6. Examine the inside of the jejunum by looking through a longitudinal incision made through its wall in the antimesenteric side. The mucosa is thrown into numerous circular folds known as the plicae circularis ().
7. Turn to the mesentery of the jejunum. By holding it stretched between your two hands and placing it against a light source you will notice that it is not fully transparent due to its content of fat and other structures (). The fat usually extends from the root of the mesentery towards the mesenteric border of the jejunum. The fat overall is less than that of the ileal mesentery.
8. The mesentery contains the jejunal branches of the superior mesenteric artery. These have numerous connections between them on their way towards the wall of the bowel (). These connections are in the form of one to three rows of small arterial arches and hence are called arterial arcades (). From the arcades spring straight arteries called the vasa recta, which are relatively long ().

STATION: SMALL INTESTINE - ILEUM

The ileum is inferior and on the right of the infracolic compartment (). There is no clear demarcation between the jejunum and ileum. The following are general features of the ileum:

1. The wall is thinner than the jejunum ().
2. The inner surface, when seen through a similar incision to the jejunum, has fewer circular folds (). Why?
3. Pick up the mesentery and examine it in the same way as in no. 7 above and compare it to that of the jejunum. It is more opaque () as it contains more fat that reaches up to the wall of the bowel and partly overlaps it ().
4. The vessels in the mesentery have more arterial arcades, about three to five rows (). The vasa recta are thus shorter ().

5. The small intestine also has lymphoid aggregations in its wall in oval patches located towards the antimesenteric border. These are known as Peyer's patches but are difficult to find in a cadaveric specimen.
6. The small intestine eventually joins the medial aspect of the caecum to form the ileocecal junction () which is guarded by the ileocecal valve formed by the slight protrusion of the terminal ileum into the cavity of the caecum. Verify this by palpating the ileocecal junction and inspecting it through an opening made in the cecal wall.

STATION: LARGE INTESTINE

The large bowel or colon forms an incomplete border around the small intestine. The direction of flow of its contents is from the cecum (blind end), with the vermiform appendix, ascending colon, transverse colon, descending colon, sigmoid colon, and finally the rectum. The large bowel has a number of features common to most of its parts. Firstly, its wall is thrown into a number of folds called sacculations or haustrae (). These are caused by the longitudinal muscle layer of the colon being shorter than the overall length of the bowel, which causes the wall to be plicated. The longitudinal muscle layer is compacted in three longitudinal bundles known as taeniae coli () that run all the way from cecum to the rectosigmoid junction (). These are found as two at the sides of the bowel and one in its posterior wall (). Along these are attached numerous fat tags termed appendices epiploicae ().

Examine these in the cadaver:

1. **The caecum** is a dilated sac with a relatively smooth wall (). It is situated in the right iliac fossa over the iliacus muscle (). Its medial border is attached to the small intestine, and just a bit further down on its posteromedial wall is the opening of the base of the vermiform appendix ().
2. The caecum is mobile but does not have a mesentery (). The peritoneum continues to cover the anterior aspect of the ascending colon and the posterior cecal wall for a variable distance before it gets reflected onto the posterior abdominal wall (). This forms the retrocecal recess, which sometimes might be very deep. This is clinically important because the tip of an inflamed appendix may lie in it.
3. The three taeniae coli of the caecum converge onto its posteromedial wall at the base of the appendix to form a complete longitudinal muscle layer for the appendix (). Verify this by following one of the cecal taeniae downwards until it leads you to the base of the appendix.
4. The vermiform appendix is a blind-ending tube about 3-5 cm in length. It is very mobile because it has a mesoappendix derived from the posterior leaf of the ileal mesentery (). The mesoappendix is triangular in shape and looks like a sail (). In its folded free margin locate the appendicular vessels (). Note that these follow the margin of the mesentery. They are also at a distance from the base of the appendix but get closer as they approach its tip (). This fact is important as the vessels may become thrombosed near the tip in cases of acute

appendicitis. This can lead to cessation of the blood circulation to the tip and perforation or gangrene with serious outcomes.

5. Note that the appendix has a complete longitudinal muscle coat and neither haustrations nor appendices epiloicae ().
6. The position of the base of the appendix is usually found by projecting a line from the anterior superior iliac spine to the umbilicus. The base lies at the junction of the lateral third and the medial two-thirds of this line (). This point is known as McBurney's point and is used to plan an appendicectomy incision.

The ascending colon runs upwards from the caecum to the right colic flexure. This is also known as the hepatic flexure of the colon. It is retroperitoneal and has all the features of the large bowel. Its blood supply comes from the medial aspect from the right colic and the ileocolic arteries of the superior mesenteric artery. They may still be covered by a peritoneal layer.

The transverse colon extends from the right to the left (splenic) flexure. It is mobile on its mesentery and has the greater omentum attached to its anterior surface (). Its blood supply is mainly from the middle colic artery of the superior mesenteric (). It runs in its mesentery usually not in the midline (). This makes a large area in the transverse mesocolon that contains no blood vessels and could be exploited surgically to gain access into the lesser sac. The left one-third of the transverse colon is the beginning of the hindgut, which receives blood supply from the inferior mesenteric artery in the form of the left colic artery and its branches.

The descending colon is very similar to the ascending colon. Distally it becomes continuous with the mobile sigmoid colon (which has a mesosigmoid and is lying intraperitoneal) that is supplied by sigmoid arteries (). The sigmoid becomes the rectum opposite the second sacral vertebra.

STATION: RECTUM

The rectum is the last part of the large bowel. It is adapted for the temporary storage of fecal matter till the time comes for its discharge. It starts as a continuation of the sigmoid colon beyond the second sacral vertebra. On the specimen verify the following:

1. The rectum has no taeniae coli, but instead a complete longitudinal muscle layer (). Likewise, there are no appendices epiloicae.
2. The rectum, when seen anteriorly, is not straight but makes side-to-side twists that cause permanent folds in its mucous membrane. These are found twice on the right side and once on the left (). From the right, the rectum follows the concavity of the sacrum and joins the anal canal at an acute angle known as the anorectal ring or junction (). This is palpable during rectal examination.
3. The peritoneum covers the upper third of the rectum anteriorly and laterally (). It covers only the front of the second part and leaves the third part without a peritoneal covering ().
4. The rectum is related posteriorly to the piriformis muscle and the sacral plexus (). Anteriorly it is related to the rectovesical pouch in the male and the

rectouterine pouch (of Douglas) in the female. The distal part of the rectum is related to the prostate anteriorly and this is useful for prostate gland examination performed through rectal examination.

STATION: ANAL CANAL

The anal canal is the last part of the GIT and represents its opening to the external environment through which the waste products are finally disposed of. In the sagittal section through the pelvis and perineum of the specimen, examine the anal canal and verify the following:

1. The anal canal is short, measuring about five centimeters in length (). Superiorly it is demarcated from the rectum, which meets with it at the anorectal angle or ring (). This demarcation shows the rectum to be wide while the anal canal is narrow. Also, the anal canal has columns while the rectum does not. The anal ring can be felt as a ridge during digital rectal examination and it can be visualized during endoscopic examination. The anorectal junction forms an angle whose opening faces posteriorly (). This is due to the pulling action of the puborectalis part of the levator ani on the junction. This also contributes to the ridging of the junction. Another factor is the presence of the internal and external anal sphincters.
2. Identify the external anal sphincter () as it forms a relatively dark layer of circular muscle around the anorectal junction. It is composed of several layers, two of which are around the anorectal junction and one that is subcutaneous and may be identified as dark fibers scattered in the subcutaneous tissue of the lower opening of the anal canal. This is a voluntary sphincter made up of striated muscle.
3. Identify the internal anal sphincter (), which is paler than the external one. This is a hypertrophied circular muscle of the wall of the anorectal junction. It is an involuntary sphincter formed of smooth muscle.
4. Turn your attention to the cavity of the anal canal and note that it is divided into an upper and lower part by the dentate line (). The upper half of the anal canal has a number of folds known as the anal columns (). These are caused by submucosal veins that anastomose between the superior and the middle and inferior rectal veins, i.e. superior to this level is drainage via the portal system, inferior to this level is drainage via the systemic circulation.
5. The columns are separated by shelves of the mucous membrane known as the anal valves (). These have small sinuses between them and the wall of the anal canal ().
6. Lower down, the anal canal ends in the anal verge which has corrugated skin (). This gradually changes color from light to dark (). This opening is the anus. It is hair bearing ().

STATION: LIVER

The liver is the largest internal organ in the body. It occupies most of the right hypochondrium and extends to the epigastric and left hypochondriac regions. Verify this in the cadaver (). It overlaps the stomach, esophagus, and structures in the posterior abdominal wall like the aorta and inferior vena cava (). It has a smooth convex diaphragmatic surface and an irregular, mostly concave visceral surface () separated from each other by the inferior border of the liver (). The diaphragmatic surface faces the anterolateral abdominal wall but is separated from it by the costal margin and the diaphragm. To this surface is attached the falciform ligament () with the obliterated umbilical vein, now known as the round ligament of the liver (ligamentum teres) in its free lower margin (). Protruding down slightly beyond the visceral surface is the fundus of the gall bladder (). The falciform ligament demarcates the anatomical right lobe from the left one (), the right being larger than the left.

The visceral surface of the liver is related to many organs. On the surface of the right side there is a deep concave impression caused by the right kidney (). Just posterior to this impression is a triangular one reaching the inferior vena cava. This is the impression of the right suprarenal gland (). Anterior to the renal impression is the shallow depression caused by the hepatic flexure of the colon (). To its left is the duodenal impression () and the gall bladder fossa in which sits the gallbladder (). On the left visceral lobe surface is a large gastric impression () with the esophageal impression leading to the gastric impression followed to the right with the pyloric impression left of the gall bladder ().

The ligamentum teres continues in a groove of the same name () till it reaches the hilum of the liver. From here, there is another groove towards the inferior vena cava. This is the groove for the ligamentum venosum (), a small fibrous cord. To the right of these two grooves are two small lobes of the liver, the caudate superiorly () near the inferior vena cava and the quadrate inferiorly near the gall bladder (). The plane that divides the liver functionally into the right and left lobes runs from the right side of the gall bladder toward the right side of the inferior vena cava.

The hilum of the liver contains the hepatic artery on the left and it can be seen branching into right and left branches (). The common bile duct lies on the right side of the artery (). It is formed from the union of the common hepatic duct with the cystic duct (). Posteriorly lies the portal vein, which has the widest lumen of the three (). The liver is covered by peritoneum on all aspects except posterosuperiorly where it is in direct contact with the diaphragm (). This is called the bare area of the liver. The layers of the peritoneum that surround this area are designated the coronary ligament. It is described as having an anterior and a posterior leaf and forming two triangular lateral ligaments ().

The gallbladder lies in its fossa on the visceral surface. It is composed of a fundus, body, neck, and cystic duct (). The cystic duct joins with the common hepatic duct to form the common bile duct. The cystic artery, a branch from the right hepatic artery, may be seen coursing towards the gallbladder. An important descriptive triangle is

formed by the common hepatic duct, cystic duct, and the surface of the liver that usually has the cystic artery running across it (). This is known as Callot's triangle. The trajectory of the inferior vena cava ends in either a groove or a tunnel into the liver (). By looking into the lumen of the IVC you may be able to see the openings of the hepatic veins, which are three to five in number ().

STATION: SPLEEN

The spleen is the largest internal lymphoid organ in the body, which serves as a biological and physical filter of the blood. It is situated in the left hypochondrium posterior to the midaxillary line () under the cover of the diaphragm. It lies parallel to the long axis of the ninth, tenth, and eleventh ribs (). It has a smooth convex diaphragmatic surface () and a concave irregular visceral surface () having the hilum in its center (). It has a sharp anterior border that is usually notched (). It is completely covered by the peritoneum but is not mobile () because it is held in place by the gastrosplenic ligament (), the lienorenal ligament (). The phrenicocolic ligament connects the left colic flexure to the diaphragm (). Its visceral surface has depressions caused by the stomach just above the hilum of the organ (). Anterior to the hilum is the area for the tail of the pancreas () and below the two is the left renal impression posteriorly and anteriorly the left colic flexure ().

Sometimes there might be small accessory spleens. Look for these near the hilum of the spleen, in the greater omentum, or the gastrosplenic omentum. They have the same color as the original spleen and look like rounded nodules. Remember there is usually one spleen, measuring 3 by 5 inches, 7 ounces, located at ribs 9 to 11 on the left side.

STATION: PANCREAS

The pancreas is one of the major exocrine glands in the body and is an accessory organ of the GIT (reminder: it is also an endocrine gland for insulin). The pancreas is retroperitoneal and fixed to the posterior abdominal wall at the level of the transpyloric plane. In the cadaver note the following:

1. The organ has a head (), neck (), body () and tail (). The head has the uncinate process extending backwards from its inferior aspect (). Note that the tail of the pancreas reaches the hilum of the spleen in the splenorenal ligament ().
2. The duodenum forms most of the relations of the head: the first part is superior to it (), the second part is on its right side (), the third part is below it (), and the fourth part, with the DJ flexure, forms the left boundary of the uncinate process ().
3. The pancreas itself lies on the commencement of the portal vein () just behind its neck
4. Posterior to the portal vein, from right to left, the pancreas runs over the inferior vena cava (), abdominal aorta (), and the left kidney ().

5. Between the lower border of the head and the upper border of the uncinate process emerge the superior mesenteric vessels ().
6. Between the head and the duodenum you may be able to locate branches of the superior pancreaticoduodenal and inferior pancreaticoduodenal vessels. These are branches of which arteries and can function as an anastomosis?
7. Look for the minor and major pancreatic duct. Which duct joins with the major pancreatic duct?
8. Look for the large artery crossing the uncinate process and the third part of the duodenum.
9. Observe the splenic artery running at the superior border of the pancreas towards the spleen and the splenic vein posterior to the spleen joining the superior mesenteric vein to continue as the portal vein. Where does the inferior mesenteric vein drain into?